



Tailoring the microstructure evolution in lunar soil simulant via thermal treatment

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ABSTRACT

Establishing lunar bases, an essential step for future space exploration, necessitates high-performance and durable construction materials. *In-situ* resource utilization (ISRU) of lunar soil to fabricate glass-ceramics offers a sustainable route to reduce dependence on costly Earth-based supply chains, contingent on precise crystallization control, a capability that critically depends on resolving key microscale dynamic features during thermal treatment. Here, we investigate the microscale crystallization behavior of a lunar soil simulant under three thermal schemes using the single hot thermocouple technique, aiming to elucidate strategies and fundamental mechanisms for controlled crystallization. Crystalline phase evolution and growth kinetics are quantitatively analyzed to reveal the interplay between nucleation and growth. Results show a transition from equiaxed to columnar and planar crystal growth as the isothermal temperature increases from 1000 °C to 1200 °C. Lower temperatures favor nucleation through stronger supercooling, while higher temperatures enhance growth via atomic mobility, resulting in up to a 28-fold variation in crystallization time. Kinetic analysis indicates volume nucleation with an activation energy of 461.55 kJ·mol⁻¹. These findings demonstrate that controlling supercooling during thermal treatment is critical for tailoring microstructures, providing guidance for the sustainable production of high-performance materials from extraterrestrial resources.

1. Introduction

Space exploration is increasingly recognized as a long-term and critical endeavor for the future of humanity [1]. The Moon, as Earth's only natural satellite and its closest celestial neighbor [2], offers a strategic location for supporting deep space missions, as its unique position makes it an ideal stepping-stone for future exploration beyond Earth. In this context, establishing lunar bases is considered essential for long-term habitation and exploration, spacecraft maintenance and refueling, and potential large-scale space migration. Such infrastructure, however, requires substantial amounts of construction materials [3]. To reduce reliance on costly Earth-based supply chains, *in-situ* resource utilization (ISRU) of lunar soil has emerged as a promising and sustainable strategy for the aspirational goal of building lunar infrastructure [4]. Given the harsh environment of Moon surface, characterized by low gravity, micrometeorite impacts, vacuum conditions, extreme temperature fluctuations, a tenuous atmosphere, and intense radiation, these materials must meet strict mechanical, thermal, and optical

standards to ensure safety and structural stability of lunar bases [5].

Currently, a broad spectrum of lunar soil-based material routes has been explored, including sintered bricks [6], geopolymers [7], ceramics [8], and glass-ceramics [9]. Within this landscape, glass-ceramics stand out as multifunctional materials with properties suitable for versatile applications, including construction components [10,11], such as walls, roads, and windows. The intrinsic hybrid structure of glass-ceramics, which comprises both glassy and crystalline phases, provides a broad spectrum of tunable properties, enabling them to better satisfy the functional requirements of lunar infrastructure compared with alternative material routes. Consequently, extensive efforts have been devoted to tailoring the optical, mechanical, and thermal properties of lunar soil-derived glass-ceramics [12–14].

The rationale for this approach is largely attributed to the chemical characteristics of lunar soil, which contains high concentrations of essential chemical compounds (e.g., CaO, SiO₂, Al₂O₃, MgO, and FeO) that collectively account for over 90% of its bulk composition, along with trace components such as TiO₂, Na₂O and K₂O, as identified in

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samples returned by the Chang'E, Apollo, and Luna missions [15,16]. This chemical composition characteristics corresponds to one of the well-established $\text{CaO-SiO}_2\text{-Al}_2\text{O}_3\text{-MgO}$ glass-ceramic systems [17], making lunar regolith a promising feedstock for *in-situ* glass-ceramic production. Owing to the extremely limited availability of actual lunar soil on Earth, current studies rely exclusively on lunar soil simulants with comparable compositions to evaluate the feasibility of glass-ceramic fabrication.

The production of glass-ceramics involves two essential stages: raw material vitrification followed by regulating crystallization. After the formation of the parent glass, the successful conversion of lunar soil into high-performance glass-ceramics primarily depends on precise regulation of the subsequent crystallization process, given that the compositional variability of lunar soil generally falls within a relatively consistent range. For instance, the formation of crystalline phases at nano-scale within the parent glass can reduce the refractive index difference to below 0.01, thereby enhancing optical performance [18]. Similarly, the development of a dense microstructure, which contributes to improved mechanical strength, is closely tied to the design of the thermal treatment regimen [19]. Under these conditions, the heat treatment schedule becomes a key factor for the desired material properties with precise control over the post-melting crystallization process.

To fully exploit this potential, it is essential to understand how post-melting thermal treatment influences the crystallization behavior of vitrified lunar soil. Key dynamic features at the microscale such as crystal shape, size, number, growth mode, and front growth rate provide valuable insight into the mechanisms that govern phase evolution [20]. Such knowledge is critical for designing heat-treatment strategies that enable precise control of microstructure and, consequently, material performance. However, microscopic investigations of this aspect remain scarce in studies of glass-ceramics-based resource utilization route, spanning both lunar soil and waste slag, which have predominantly relied on post-treatment experimental characterization techniques [21–23], such as XRD, DSC, Raman spectroscopy, FTIR, and SEM. While these methods provide valuable insights into the final material properties at the macroscopic scale of mass transfer behavior, they fail to capture key crystallization parameters that illustrate the dynamic relationship between heat treatment conditions and crystal growth, which is essential for tailoring the microstructure evolution over the crystallization process and optimizing material performance.

In studies of crystallization-driven phase transformations, three representative techniques are commonly used to probe crystallization behavior under controlled thermal conditions: directional solidification technique (DST) [24,25], confocal laser scanning microscopy (CLSM) [26,27] and the single hot thermocouple technique (SHTT) [28,29]. While DST enables simultaneous tracking of thermal history and phase evolution, it lacks the capability to resolve real-time microstructural dynamics. In contrast, direct and real-time observation of crystallization under controlled thermal conditions offers an effective way to probe these key dynamic features at the microscale. CLSM and SHTT provide *in-situ* observation, among which SHTT has emerged as a particularly versatile approach due to its flexible thermal control (cooling rates up to $150\text{ }^\circ\text{C}\cdot\text{s}^{-1}$) and its ability to resolve microscale crystallization dynamics [30].

In summary, ISRU of lunar soil to produce high-performance glass-ceramics offers a promising pathway to reduce dependence on costly Earth-based supply chains for extraterrestrial construction, provided that the crystallization process can be precisely controlled, a capability that critically depends on resolving key microscale dynamic features. However, the effects of thermal treatment temperature and duration on this microscale dynamic crystallization process remain poorly understood, motivating a systematic and well-controlled experimental investigation. Herein, lunar soil simulants were likewise employed to investigate microscale crystallization behavior under systematically varied heat treatment conditions, specifically focusing on temperature

and duration. Using *in-situ* observation via sophisticated equipment (i.e., SHTT), we examined how thermal parameters influence dynamic process of nucleation and crystal growth, aiming to identify strategies and fundamental mechanisms for precise crystallization control. The results are expected to enhance the fundamental understanding of crystallization mechanisms and provide practical guidance for the fabrication of high-performance lunar soil-derived glass-ceramics, all of which contribute to efficient resource management and sustainable extraterrestrial-terrestrial resource cycle.

2. Methodology

2.1. Experimental system and procedure

A real-time observation of crystallization phenomena requires precise control of thermal treatment conditions and the ability to monitor structural changes at the microscale. The SHTT experimental system is designed to meet this objective, consisting of three main components: the SHTT embodiment, a temperature controller and a PC, as shown in Fig. 1(a). The SHTT setup comprises a camera, a microscope, and a furnace chamber, arranged vertically from top to bottom.

The furnace chamber is equipped with four thermal radiation lamps at the bottom that provide uniform illumination for clear observation at temperatures below $1000\text{ }^\circ\text{C}$, while also enhancing heating capacity at higher temperatures (particularly above $1400\text{ }^\circ\text{C}$) by maintaining an ambient environment of $600\text{--}700\text{ }^\circ\text{C}$. Inside the chamber, a type-B thermocouple (PtRh30-PtRh6, diameter of $\phi = 0.5\text{ mm}$) holds the simulant sample. The thermocouple serves a dual function: acting as both the heating element and the temperature sensor, enabling precise thermal regulation with an accuracy of $\pm 1\text{ }^\circ\text{C}$, as the sample is shaped similarly to a grain of rice. The temperature controller governs the heat treatment schedule, while the connected PC facilitates the input of programmed thermal cycles and continuously records the crystallization process in real time (Recorded at 1280×890 pixels, corresponding to a spatial resolution of 0.002 mm per pixel, with a temporal interval of 0.25 s between frames).

Before conducting the SHTT tests, the prepared samples were first melted at $1500\text{ }^\circ\text{C}$ for 60 min , followed by rapid water quenching to ensure the formation of fully amorphous glass (parent glass). The quenched samples were then dried and ground to a particle size below 200 mesh . Three thermal treatment schemes were subsequently applied, as illustrated in Fig. 1(b), to mimic realistic glass-ceramic processing conditions. All schemes shared the same initial heating protocol: a complete remelting at $1500\text{ }^\circ\text{C}$ for 60 s with a heating rate of $15\text{ }^\circ\text{C}\cdot\text{s}^{-1}$, followed by rapid cooling at $200\text{ }^\circ\text{C}\cdot\text{s}^{-1}$ before initiating the SHTT observation.

Schemes I and II were designed to investigate the effect of isothermal temperature on crystallization, using setpoints of 1000 , 1100 , 1200 , and $1300\text{ }^\circ\text{C}$. In Scheme I, the temperature was directly reduced to each target isothermal value. In contrast, Scheme II included a 4-second hold at $1200\text{ }^\circ\text{C}$ immediately after cooling, followed by a temperature increase or decrease (excluding $1200\text{ }^\circ\text{C}$) at a rate of $80\text{ }^\circ\text{C}\cdot\text{s}^{-1}$ to reach the target isothermal condition. Samples in Scheme III were initially cooled to either $950\text{ }^\circ\text{C}$ or $1100\text{ }^\circ\text{C}$ and then switched to the alternate temperature ($1100\text{ }^\circ\text{C}$ or $950\text{ }^\circ\text{C}$, respectively). For both transition directions, the samples were first soaked for 5 , 10 , or 20 s and then underwent the temperature transition to assess the effect of varied thermal durations on crystallization behavior. The rationale behind this experimental selection is relevant to cooling rate, with a detailed explanation provided in the Supporting Information.

2.2. Simulated lunar soil

To date, lunar soil samples have been successfully returned by the Chang'E, Apollo, and Luna missions, with each sample collected from different lunar sites exhibiting distinct chemical compositions. Notably,

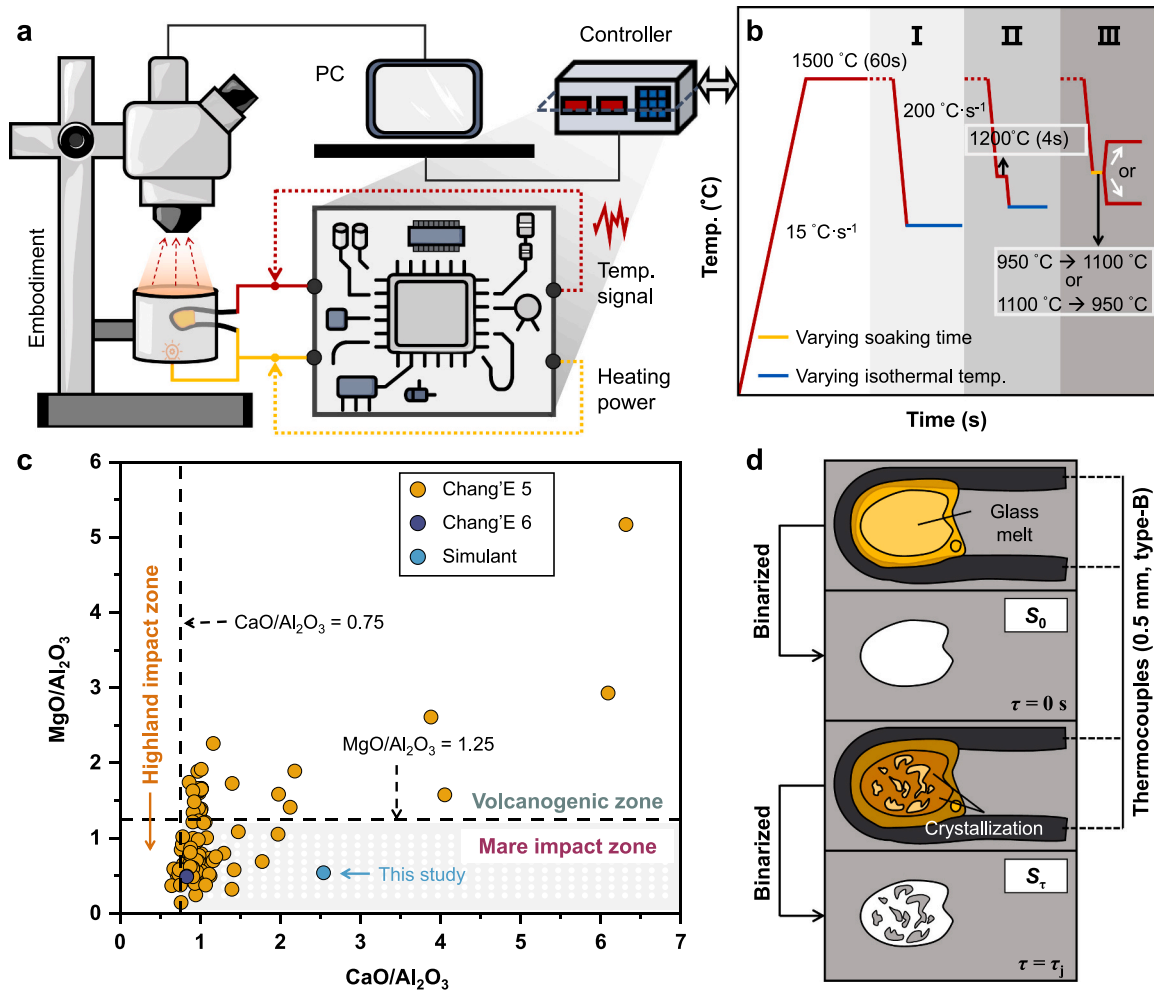


Fig. 1. Methodological framework of this study. (a) Overview of the SHTT experimental system. (b) The three designed thermal cycling methodologies, each sharing the same initial heating process prior to the SHTT tests. (c) Compositional profiles of returned lunar soils, compiled using the simulant of this study and the latest chemical composition datasets for the Chang'E missions (see [Supporting Information](#)). (d) Image processing workflow used in the SHTT tests to analyze the crystallization dynamics of simulated slag materials.

the Apollo 15 and 17 samples were retrieved from junction zones between the mare plains and the highlands, whereas other samples were collected exclusively from either highland or mare plain regions. Moreover, the Chang'E 6 sample is the most recently obtained and the only one sourced from the Moon's far side, which provides valuable insight into the longstanding compositional dichotomy between the nearside and far side of the Moon from a sampling perspective.

Importantly, the selection of future lunar base sites should prioritize flat regions within the mare plains, as such terrains offer reduced construction complexity and shorten time costs for establishing a stable operational environment. Lunar soil samples with differences in geographic origin and terrain share a consistently high SiO₂ content, along with notable concentrations of CaO, Al₂O₃, MgO and FeO [31]. Compositional profiles, however, can help distinguish between soils of highland and mare origin. In particular, the formation of glasses from meteorite impacts in these regions produces distinct chemical signatures: samples with MgO/Al₂O₃ < 1.25 and CaO/Al₂O₃ > 0.75 are generally indicative of mare-derived materials [32,33], as shown in Fig. 1(c).

Since the formation of a stable parent glass is the first critical step in glass-ceramics production, a representative simulant was formulated using reagent-grade powders, primarily targeting the key oxides of lunar soil, with details presented in [Supporting Information](#). The composition of the prepared simulant falls squarely within the mare-influenced range in Fig. 1(c), given the natural variability in lunar soil chemistry.

Although the experimental conditions differ from the actual lunar environment, particularly in terms of gravity, this simplification is justified within the scope of the present study. The experiments focus on the dynamic evolution of crystalline phases at the microscale using a very small simulant sample (approximately $\sim 10^{-3}$ g). The influence of buoyancy-driven convection induced by gravity on crystallization can be assessed using the Grashof number (Gr), where $Gr \gg 1$ signifies buoyancy effects-dominated transport [34]. Here, Gr is estimated to range from 0.142 to 0.249 based on simplified assumptions (see the [Supporting Information](#) for details), suggesting that buoyancy-driven convection is reasonably negligible, and that crystallization behavior is instead governed primarily by chemical composition characteristics and the applied heat-treatment schedule. In addition, the small sample size allows precise control of heating/cooling rates, enabling accurate observation of intrinsic crystallization mechanisms.

2.3. Data processing

A series of time- and temperature-labeled snapshots (τ , T) were obtained following each SHTT test, enabling the extraction of crystallization parameters from a microscopic perspective. Key parameters such as the incubation time (τ_i , defined as the time required for nucleation at a specific isothermal temperature) and the overall crystallization time (τ_g , defined as the time interval from τ_i to the end of crystallization) can be directly identified first. Utilizing ImageJ software, we further estimated

the crystal phase content (α) at a given moment τ during crystallization based on the optical contrast between the transparent molten slag and the opaque crystallized regions, as shown in Fig. 1(d). The process involved binarizing the initial image at $\tau = 0$ s to define the liquid slag area (S_0), extracting the profile of the crystallized region (dark yellow) at τ , and filling it with pixels. The crystallized area (S_τ) was then quantified by pixel count, and α was calculated as the ratio of crystallized area to initial liquid slag area (S_0):

$$\alpha(\%) = S_\tau / S_0 \quad (1)$$

At the end of the crystallization process, α reaches 100%, allowing us to derive the average crystal growth rate (v_a) over the entire crystallization period and the instantaneous crystal growth rate (v_τ) at τ . They

can be estimated using the following equations:

$$v_a (\% \cdot s^{-1}) = 100\% / \tau_g \quad (2)$$

$$v_\tau (\% \cdot s^{-1}) = d\alpha / d\tau \quad (3)$$

Meanwhile, ImageJ software was also employed to measure the diameter (d) and number (N) of crystal grains, as well as the translational velocity (v) of the crystal growth front at various growth stages and modes.

In terms of uncertainty analysis, the maximum temporal error for τ_i and τ_g was limited to 0.25 s, which required a frame capture rate of four images per second. However, it is important to note that the optical appearance of the opaque regions may not represent a fully crystallized

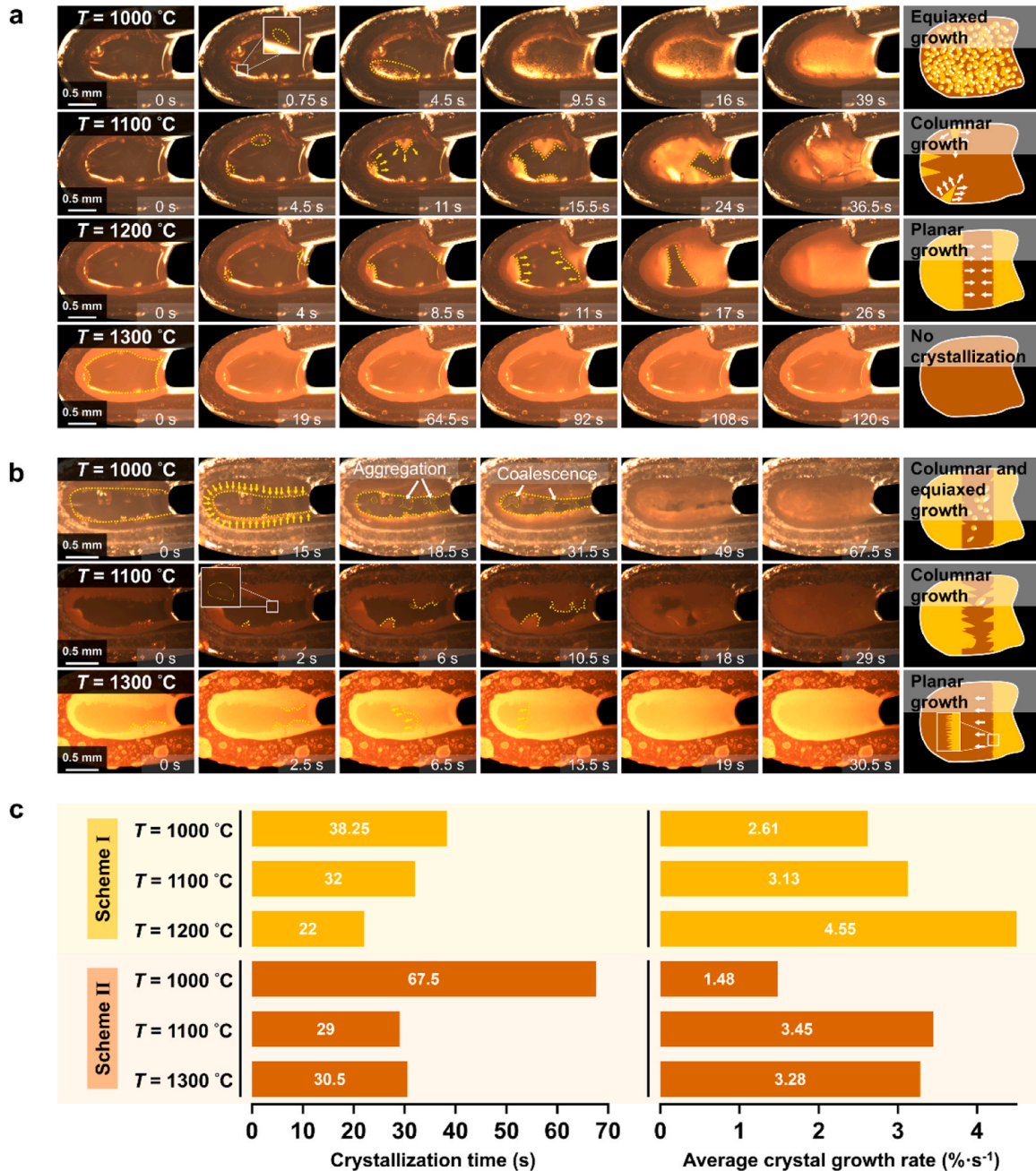


Fig. 2. Microscopic examination of crystallization behavior. Sequential snapshots of the simulant slag at isothermal conditions for (a) Scheme I and (b) Scheme II, accompanied by schematic illustrations of the corresponding crystal growth modes shown on the right. The time label “0 s” denotes the onset of the target isothermal temperature. (c) Dynamic related-parameters: the crystallization time and the corresponding average crystal growth rates. For Scheme II, the crystallization time was measured from 0 s, as the incubation stage occurred concurrently with the temperature transition.

phase (100%), especially at $T > 1200$ °C, introducing potential uncertainty in the estimation of α and v_c . We provide a reproducibility assessment in the [Supporting Information](#), showing that the experimental results demonstrate good reproducibility and an acceptable level of precision.

3. Results and discussion

3.1. Phenomenological crystallization behavior

3.1.1. Effect of Scheme I and Scheme II

Following the SHTT tests, sequential snapshots were examined to track the crystallization behavior of the lunar soil simulant during thermal treatment. This microscopic examination provided a direct and time-resolved view of phase evolution under different thermal conditions. [Fig. 2](#) compares the results for Scheme I and Scheme II, presenting both the visual sequence of crystallization events and the associated dynamic parameters.

In [Fig. 2\(a\)](#), the molten slag appears as a transparent liquid at the moment when cooling at the target isothermal temperature (time = 0 s). Crystal phase precipitation generally requires an initial period for nucleation site formation, observed within the first few seconds as sites emerged either at the slag-thermocouple (or air) interface or within the melt. This incubation time (τ_i) was 0.75 s, 4.5 s, and 4 s at $T = 1000$ °C, 1100 °C, and 1200 °C, respectively, showing an overall increase with temperature. The trend reflects the reduced degree of supercooling at higher isothermal temperatures, which lowers the probability of nucleation and prolongs the incubation stage. Notably, the degree of supercooling was insufficient to trigger nucleation at $T = 1300$ °C, and no crystallization was observed within 120 s.

Crystal growth patterns changed markedly with temperature in [Fig. 2\(a\)](#). At $T = 1000$ °C, numerous fine and spherical crystals aggregated in the lower-left region before spreading evenly across the slag. At $T = 1100$ °C, dendritic or zigzag crystals developed from the slag-thermocouple interface toward the center, producing a clear phase boundary by the end of crystallization (36.5 s). Ultimately, strip-like crystals formed simultaneously from the left and right edges at $T = 1200$ °C, advancing toward the center and likewise creating a distinct phase boundary after solidification (complete opacity).

The crystal growth patterns observed in [Fig. 2\(a\)](#) can be classified as equiaxed growth at 1000 °C, columnar growth at 1100 °C, and planar growth at 1200 °C. These modes are influenced by changes in supercooling degree, slag viscosity, and heat flux direction [20]. Since these factors are governed by the thermal treatment conditions, careful design of the heating scheme is essential for achieving precise control over crystallization in lunar soil-based glass-ceramics. As demonstrated in [Fig. 2\(a\)](#), varying the isothermal temperature not only determined whether crystallization occurred but also influenced crystal morphology, both of which directly affect the final material properties.

Scheme II was used to further examine how a tailored cooling path influences crystallization, as shown in [Fig. 2\(b\)](#). This approach involved a temperature-jump process: rapid cooling at 1200 °C for 4 s (equal to the incubation time) followed by a shift to the target isothermal temperatures of 1000 °C, 1100 °C, or 1300 °C. After the temperature jump, a mixed growth mode combining equiaxed and columnar structures was observed at $T = 1000$ °C, with equiaxed growth more pronounced during cooling process. The formation of equiaxed crystals was marked by the aggregation and complete coalescence of adjacent grains over time. At $T = 1300$ °C, crystallization that was absent under direct isothermal holding occurred after the temperature jump due to the now favorable degree of supercooling. In this case, growth proceeded primarily from right to left in a planar growth mode, producing a cloudy appearance in the solidified process.

When the degree of supercooling is sufficient and the growth direction aligns with the heat flux, dendritic or zigzag crystals typically form throughout the crystallization process, hallmarking of columnar growth.

In contrast, increased molten slag viscosity acts as a resistance to crystal growth, favoring the development of spherical crystals characteristic of equiaxed growth. If the supercooling degree is inadequate, growth proceeds in a direction opposing the heat flux. In this case, local nucleation at the crystal-melt interface is not triggered, and crystallization advances only as a planar front.

[Fig. 2\(c\)](#) shows the crystallization time (τ_g) and average crystal growth rate (v_a) for each growth mode under the two heat treatment schemes. In Scheme I, τ_g decreased with isothermal temperature, producing an inverse trend in v_a . At $T = 1000$ °C, τ_g reached 38.25 s, the longest among the tested temperatures in Scheme I, while τ_i was shortest at 0.75 s. This suggests that lowering the isothermal temperature promotes a higher degree of supercooling, which facilitates nucleation, but also increases slag viscosity, mitigating molecular diffusion and crystal growth. The result is slower growth of spherical crystals characteristic of the equiaxed mode. Higher isothermal temperatures (1100 °C and 1200 °C) provided greater thermal energy for post-nucleation growth, shortening τ_g by approximately 16.34% and 42.48%, respectively, compared with $T = 1000$ °C.

In Scheme II, the temperature-jump process markedly prolonged crystallization at $T = 1000$ °C, with τ_g reaching 67.5 s (about 1.74 times longer than in Scheme I). This can be attributed to the coexistence of equiaxed and columnar growth modes, which requires greater molecular mobility while the available thermal energy remains fixed, resulting in a lower v_a of $1.48\% \cdot s^{-1}$. At $T = 1100$ °C, the 4 s pre-nucleation stage in Scheme II enabled a slightly higher growth rate of about $3.45\% \cdot s^{-1}$ than that in Scheme I, and Scheme II also exhibited a columnar growth mode. While 1200 °C in Scheme I and 1300 °C in Scheme II both produced planar growth, the higher treatment temperature in Scheme II generated a pronounced re-melting zone at the growth front. This created a mushy region between the crystal phase and liquid slag, restricting growth and extending τ_g by 8.5 s compared with Scheme I.

3.1.2. Effect of scheme III

Building on the above discussion, temperature transitions in heat treatment can be used to reach controllable crystallization during cooling. To explore this further, Scheme III was designed to evaluate how varying transition conditions affect crystallization, with the results shown in [Fig. 3](#). From earlier results, the τ_i of the simulant at 1100 °C was determined to be 4.5 s. Therefore, soaking times were set beyond this value, specifically at 5, 10, and 20 s.

As shown in [Fig. 3\(a\)](#), crystallization at $T = 1100$ °C began primarily at the interface between the liquid slag and the thermocouple, with longer soaking durations producing a larger columnar crystal. After the subsequent transition to 950 °C, the pre-formed columnar crystals continued to grow slowly, while equiaxed crystals also gradually appeared within the melt. During the 1100 °C soaking stage, crystal growth followed the heat flux direction in columnar form, but after transitioning to 950 °C, the increased viscosity reduced columnar growth and simultaneously promoted equiaxed growth.

In contrast, for the 950 °C–1100 °C transition shown in [Fig. 3\(b\)](#), the slag remained transparent during the 950 °C soaking stage across all soaking times, indicating that direct isothermal holding at 950 °C required an incubation time longer than 20 s. Once the temperature transitioned to 1100 °C, however, spherical crystals precipitated immediately within the melt and rapidly expanded to occupy the whole region. This transition also resulted in a mixed growth mode of equiaxed and columnar crystals, where equiaxed growth dominated at all soaking times. An interesting observation was made at the 5 s soaking time during this transition: a large bubble appeared at the end of crystallization shown in [Fig. 3\(b\)](#). This may have resulted from rapid crystal growth expelling viscous melt toward the center, forming a film-like structure.

The occurrence of mixed growth modes can be explained by the interplay of random nucleation events during both temperature transitions, where both heterogeneous and homogeneous nucleation were

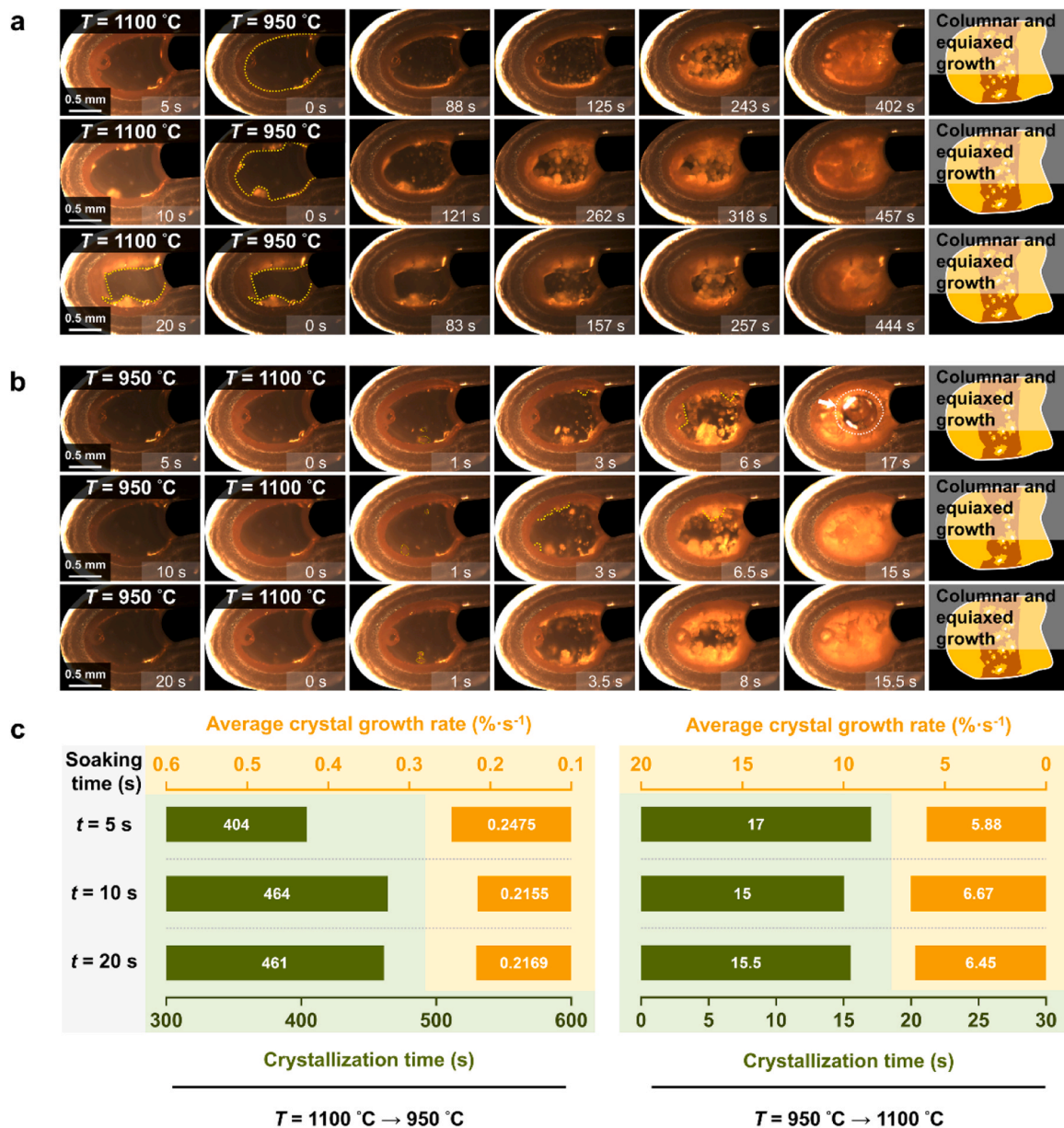


Fig. 3. Microscopic examination of crystallization behavior in Scheme III. Sequential snapshots of the simulant slag under isothermal conditions during temperature transitions from (a) 1100 °C to 950 °C and (b) 950 °C to 1100 °C, recorded at different soaking times. The time label “0 s” denotes the onset of the target isothermal temperature. (c) Dynamic related-parameters: the crystallization time and the corresponding average crystal growth rates.

possible [35]. In the temperature transition of 1100 °C–950 °C, crystal precipitation sites were predominantly located at the slag-thermocouple interface, reflecting heterogeneous nucleation. The energy barrier for crystal formation is lower at the interface [36,37], making nucleation more favorable than in the bulk melt. By contrast, in the 950 °C–1100 °C transition process, homogeneous nucleation was more evident, as the combined effect of higher supercooling and increased viscosity promoted the spontaneous formation of tiny clusters within the liquid melt. Once the temperature was raised to 1100 °C, the concurrent increase in thermal energy supply and decrease in slag viscosity facilitated rapid equiaxed crystal growth, leading to the swift occupation of the melt by crystallites.

Despite the similar growth morphologies observed under both temperature transition pathways, their crystallization dynamics differed substantially, as illustrated in Fig. 3(c). Specifically, the crystallization time in the 950 °C–1100 °C transition was nearly 28 times shorter than that of the 1100 °C–950 °C transition when averaged across the three

soaking durations. This contrast highlights the dual role of temperature: lower isothermal conditions favor nucleation due to the sufficient degree of supercooling, whereas higher isothermal conditions promote rapid crystal growth by reducing viscosity and providing the necessary thermal energy for atomic rearrangement and mobility. For the 1100 °C–950 °C pathway, extended soaking allowed crystals to precipitate and grow at $T = 1100\text{ °C}$, but the subsequent cooling to $T = 950\text{ °C}$ suppressed growth and strongly promoted secondary nucleation. Effectively, as the increase of soaking time at $T = 1100\text{ °C}$, this nucleation-dominated stage became more pronounced after the transition, which prolonged overall crystallization and yielded finer grains. In contrast, under the 950 °C–1100 °C pathway, nuclei pre-formed at $T = 950\text{ °C}$ could immediately grow once reheated, and higher soaking durations further increased the number of active nuclei, collectively leading to much shorter crystallization times.

3.2. Quantitative assessment of crystallization behavior

Comparison of the three designed thermal schemes shows that the sequence of temperature transitions plays a decisive role in crystallization behavior by shaping the balance between nucleation and growth. By deliberately adjusting the cooling path to develop either nucleation-dominated or growth-dominated regimes, the resulting microstructure can be effectively controlled. This provides a practical approach for optimizing the preparation of lunar soil-based glass-ceramics. To build on these observations, this section presents a quantitative assessment of the crystallization process. Key metrics and kinetic parameters are introduced to evaluate how different cooling transition pathways influence crystallization efficiency and microstructural development.

3.2.1. Crystallization metrics

Different crystal growth modes are associated with distinct crystallization dynamics, which can be evaluated by tracking the change in crystal phase content over time using the snapshot-based workflow. The estimated contents and corresponding growth rates for all three schemes are presented in Fig. 4. Across all cases, the crystal phase content followed a similar trend: a rapid increase during the early stage, followed by a gradual approach to 100%. This produced a hump-shaped pattern in the growth rate curves. The behavior can be explained by the availability of growth space. At the onset of crystallization, sufficient space allows rapid expansion of crystals. As growth continues, adjacent grains begin to impinge and restrict each other, reducing the rate until the remaining liquid slag disappears and crystallization is complete. This general pattern is independent of the specific growth mode.

The imposed isothermal conditions in each scheme strongly influenced both the magnitude and duration of crystal growth, as reflected in the fluctuations of growth rate over time. In all the rate curves, a single peak or two distinct peaks were observed, highlighting pathway-dependent differences in crystallization kinetics. Specifically, the peak values in Fig. 4(a) of v_r were 5.763%, 5.765%, and 9.259%·s⁻¹ at $T = 1000$, 1100, and 1200 °C, respectively, showcasing an increase with

rising temperature. In Scheme II, the growth rate curves at both 1000 and 1300 °C displayed two peaks, with the highest value of $v_r = 7.828\% \cdot s^{-1}$ at $T = 1300$ °C. Scheme III also consistently exhibited two growth-rate peaks under different soaking times, with the peak values differing by several tens to over one hundred times between the two transition pathways.

When a single growth mode occurred under a constant cooling rate, crystallization proceeded with one dominant growth rate peak, as shown in Fig. 4(a). By contrast, schemes with temperature jumps altered the cooling rate and produced varying degrees of supercooling, which in turn drove mixed growth modes and resulted in multiple peaks. A large temperature drop created a high cooling rate and strong supercooling, conditions that primarily promoted nucleation over growth. This was observed at $T = 1000$ °C in Scheme II and in the 1100 °C–950 °C pathway of Scheme III. Conversely, when the transition was to higher temperatures, the cooling rate was lower and the degree of supercooling reduced, favoring growth over nucleation. This behavior is exemplified by the 950 °C–1100 °C pathway of Scheme III. However, a notable exception was observed at $T = 1300$ °C in Scheme II, where the elevated temperature resulted in a pronounced re-melting zone at the growth front, which in turn triggered a secondary nucleation-growth cycle and led to fluctuations in the growth-rate curve. Overall, the shifting balance between nucleation and growth during the crystallization process explains the presence of single- versus multi-peak growth-rate curves across different schemes.

The preceding analysis of crystal phase content evolution offers a broad view of the overall crystallization process. However, because different thermal schemes can produce distinct growth modes, phase content and its rate of change alone are insufficient to fully characterize crystallization behavior. For instance, equiaxed growth often yields small, sphere-like grains, making it essential to analyze both crystal number and size in order to guide the development of uniform microstructures. To capture these additional aspects, we further evaluated three key crystallization metrics: the translational velocity of the crystal growth front (v), crystal diameter (d) and crystal number (N).

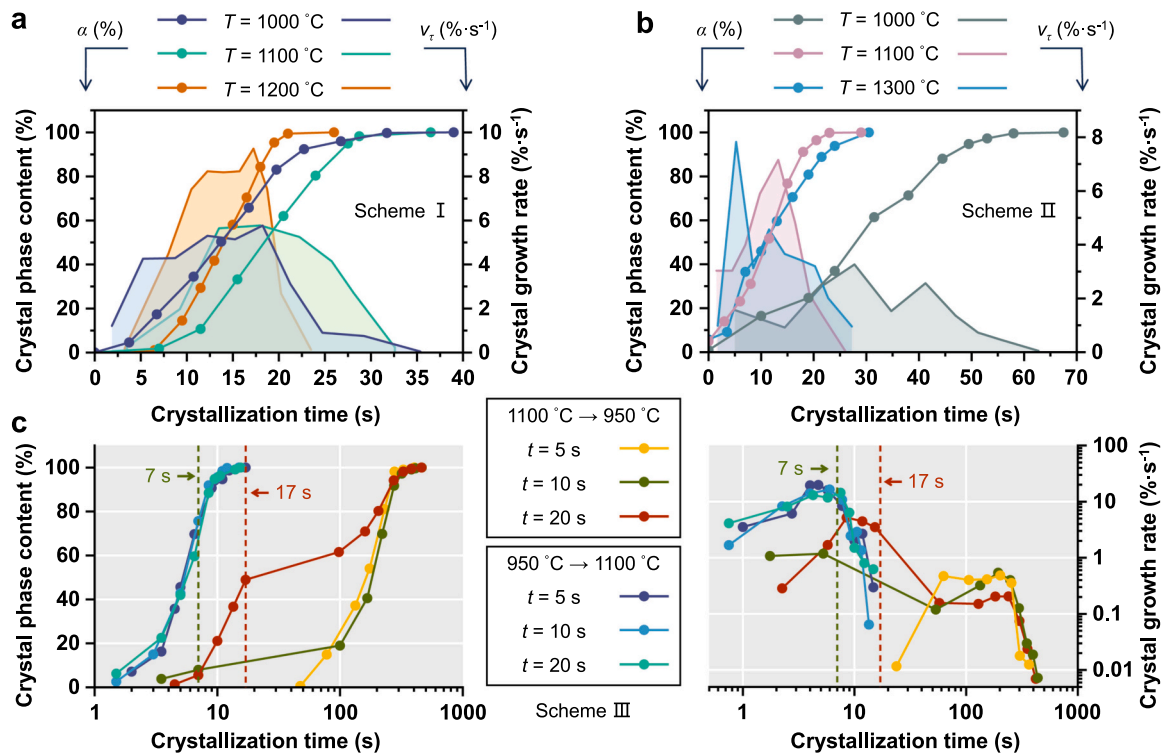


Fig. 4. Temporal evolution of the estimated crystal phase content and corresponding growth rates. (a) Scheme I. (b) Scheme II. (c) Scheme III. In (a) and (b), the filled area underneath the line chart represents the crystal growth rate.

In columnar or planar growth modes (including cases where they appeared within mixed modes), crystals typically advance from the interface between the liquid slag and the thermocouple toward the slag center at a measurable velocity. We tracked this translational rate of the growth front (defined as the fastest advancing direction) by measuring the distance from the root to the contact point, with the results shown in Fig. 5(a). During free growth, higher isothermal temperatures generally produced higher translational velocities with greater fluctuations. At $T = 1200^\circ\text{C}$ in Scheme I, v ranged between 33.846 and $64.667\ \mu\text{m}\cdot\text{s}^{-1}$, while it decreased to approximately 15.714 – $24\ \mu\text{m}\cdot\text{s}^{-1}$ at $T = 1000^\circ\text{C}$. In Scheme II, velocities decreased steadily with the reduced temperature: 17 – $33.333\ \mu\text{m}\cdot\text{s}^{-1}$, 11 – $25\ \mu\text{m}\cdot\text{s}^{-1}$, and 2 – $4.4\ \mu\text{m}\cdot\text{s}^{-1}$, respectively, with comparatively stable fluctuations. In Scheme III, the 950°C – 1100°C transition pathway yielded velocities of 6 – $28.67\ \mu\text{m}\cdot\text{s}^{-1}$ across all soaking times. In contrast, under the reverse 1100°C – 950°C transition, velocities dropped sharply to 0.015 – $14\ \mu\text{m}\cdot\text{s}^{-1}$ and remained nearly constant once the isothermal condition was changed to 950°C , reflecting the effect of increased cooling rate. These results indicate that planar growth exhibits the strongest translational ability.

In the equiaxed growth mode, crystals precipitated within the liquid slag typically displayed irregular, sphere-like shapes and occurred in large numbers, making their statistical characterization essential for

understanding crystallization behavior. Because of the irregular morphology, direct size measurement was not possible. Instead, we applied a simplified method: the area of targeted crystal was measured and then converted into the area of a circle with equivalent size, from which the effective diameter was calculated. As shown in Fig. 5(b), the average crystal diameter increased steadily over time. This resulted from agglomeration and coalescence between neighboring grains during free growth, correspondingly leading to an initial increase followed by a subsequent decrease in the number of crystals.

A temperature jump significantly affected the maximum average diameter: at 1000°C , the difference between Scheme I and Scheme II reached $\Delta = 55.649\ \mu\text{m}$ (a difference of 59.2%), where Scheme I presented a smallest average diameter = $26.543\ \mu\text{m}$. In addition, Scheme I produced a narrower size distribution, attributable to much higher precipitated crystal number (mean = $952\ \text{mm}^{-2}$, nearly an order of magnitude greater than Scheme II) within a limited growth space. Crystal diameters at different soaking times were generally comparable within the same transition pathway of Scheme III. However, the distribution range gradually narrowed with longer soaking, indicating that extended soaking favored the development of a more uniform crystal size distribution. Interestingly, the crystal number in Fig. 5(c) was comparable between the two transition pathways after the temperature

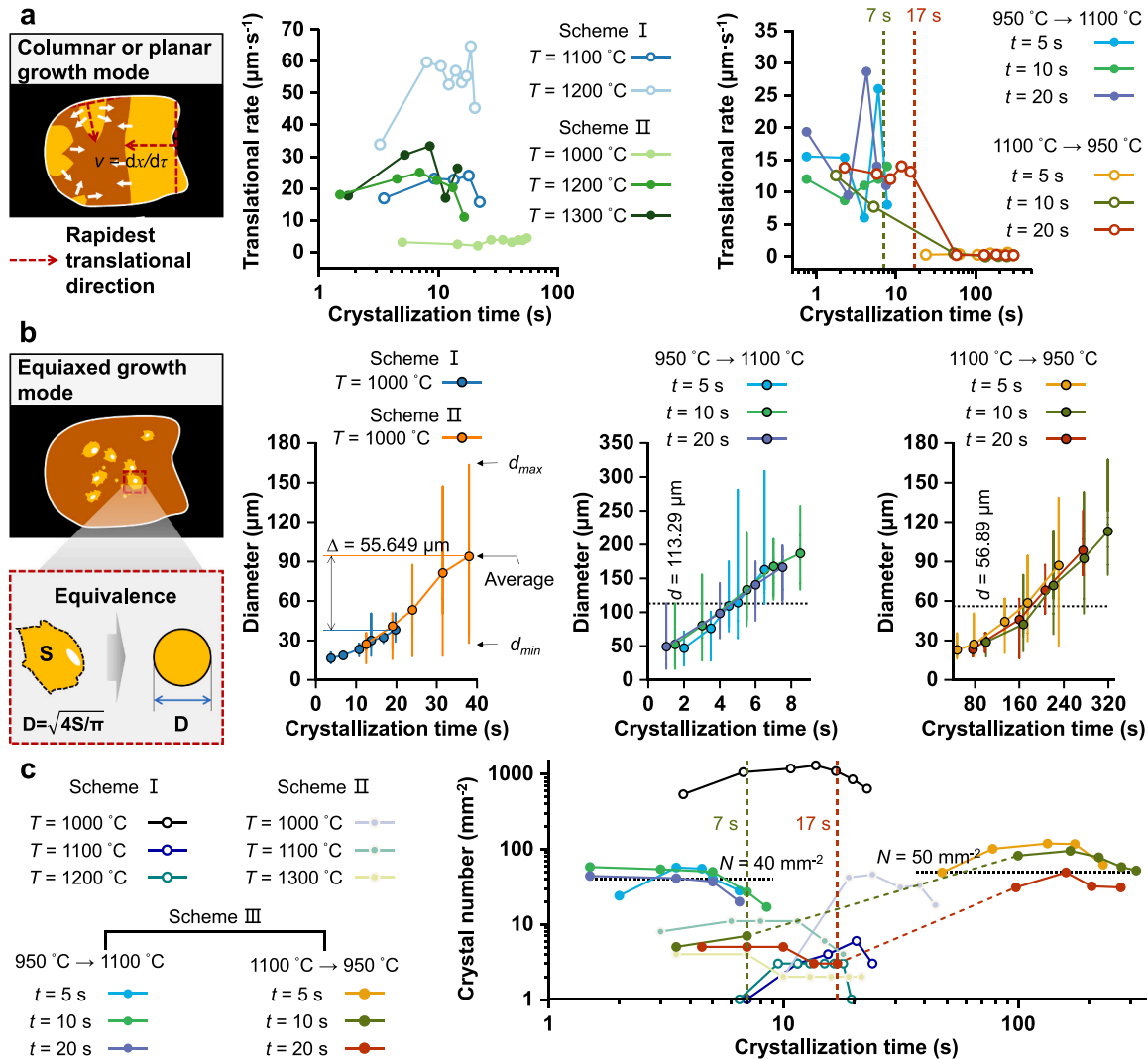


Fig. 5. Temporal evolution of crystallization metrics under different Growth modes across the designed schemes. (a) Translational rate of the crystal growth front, highlighting the fastest translational direction. (b) Crystal diameter, with vertical bars showing the distribution range ($d_{\min}$ – $d_{\max}$) and spherical symbols denoting average values (statistical count = 6), connected to illustrate the variation trend. (c) Crystal number per square millimeter. For Scheme III, $\tau - \tau_1 = 7\ \text{s}$ and $\tau - \tau_1 = 17\ \text{s}$ indicate the moments when the isothermal temperature in the 1100°C – 950°C pathway shifted to 950°C at soaking times of $t = 10\ \text{s}$ and $t = 20\ \text{s}$, respectively.

shift, fluctuating around $N = 40 \text{ mm}^{-2}$ (for $950^\circ\text{C} - 1100^\circ\text{C}$ pathway) and $N = 50 \text{ mm}^{-2}$ (for $1100^\circ\text{C} - 950^\circ\text{C}$ pathway). Despite this, the average diameters differed by nearly a factor of two, as illustrated in Fig. 5(b).

In summary, crystallization behavior under different thermal schemes is governed by the dynamic balance and competition between nucleation and growth, which collectively dictate both phase content evolution and microstructural features. Based on classical nucleation theory and crystal growth theory, the steady-state nucleation rate (I) and crystal growth rate (U) can be expressed as [17]:

$$I = \frac{2D}{a^4} \sqrt{\left(\frac{\sigma}{k_B T}\right)} e^{-\frac{\Delta G_{\max}}{k_B T}} \quad (4)$$

$$U = \frac{CT}{\eta} \left(1 - e^{-\frac{\Delta G}{k_B T}}\right) \quad (5)$$

where D is the diffusion coefficient, a is the characteristic size of a structural unit, k_B is the Boltzmann constant, σ is the interfacial energy per unit area, ΔG is the Gibbs free energy for nucleus formation (with the subscript "max" denoting the critical cluster), C is a constant associated with crystal growth parameters (e.g., the characteristic length of the structural unit and the growth mechanism), and η is the viscosity.

Fig. 6(a) schematically illustrates the steady-state variation of I and U , each reaching a maximum at a distinct temperature between the liquidus temperature (T_l) and the glass transition temperature (T_g). This theoretical trend is consistent with the observed crystallization behavior under different thermal schemes: lower isothermal temperatures promote nucleation, whereas higher temperatures favor crystal growth. The occurrence of multiple peaks in Fig. 4 can be attributed to temperature-alternating schemes temporarily disrupt the steady state of I and U .

From a heat and mass transfer perspective, the radial temperature gradient (dT/dx) at initial cooling stage governs the final crystal growth mode. During this stage, the sample traverses characteristic temperatures: T_l , crystallization onset (T_o), crystallization end (T_e), and T_g , which define liquid ($> T_l$), mushy ($T_l - T_g$), and solid ($< T_g$) regions. Crystallization predominantly occurs between T_o and T_e under slow cooling, accompanied by latent heat release [38]. As shown in Fig. 6(b), cooling from 1500°C to an isothermal temperature (T) establishes a dT/dx between the center and boundary, driving heat transfer. At lower isothermal temperatures ($T < T_e$), a large dT/dx and significant latent heat release enhance heat dissipation due to reduced thermal resistance (R_{th}). The resulting rapid viscosity increase suppresses atomic mobility,

promoting uniform, equiaxed crystal growth. In contrast, at higher isothermal temperatures ($T > T_e$), the reduced dT/dx and increased R_{th} limit heat dissipation, enabling a slow temperature decline to maintaining a lower viscosity, which facilitates atomic transport toward preferred nucleation sites, typically at the boundary (where the intense cooling occurs), leading to directional growth along the heat flux and favoring columnar or planar crystal structures.

3.2.2. Crystallization kinetics

Under isothermal conditions, the crystallization kinetics of the lunar soil simulant slag were analyzed using the Johnson-Mehl-Avrami (JMA) model [39,40], which is given by:

$$\alpha(\tau - \tau_i) = 1 - e^{-K(\tau - \tau_i)^n} \quad (6)$$

The JMA equation can be expressed in logarithmic form as:

$$\ln[-\ln(1 - \alpha)] = n \ln(\tau - \tau_i) + \ln K \quad (7)$$

where n is the Avrami exponent, reflecting the nucleation and growth mechanism, and K is the effective overall reaction rate. The value of n is indicative of the crystallization behavior [41]: $n = 4$ corresponds to volume nucleation with 3D growth, $n = 3$ to volume nucleation with 2D growth, $n = 2$ to volume nucleation with 1D growth, and $n = 1$ to surface nucleation with 1D growth. The parameter K is related to the energy barrier for crystal precipitation and follows the Arrhenius relation:

$$K = K_0 e^{(-E/RT)} \quad (8)$$

which can also be expressed in logarithmic form:

$$\ln K = -E/RT + \ln K_0 \quad (9)$$

where E is the activation energy ($\text{kJ}\cdot\text{mol}^{-1}$), R is the gas constant ($8.314 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$), T is the absolute temperature (K), and K_0 is referred as the frequency factor (which can be determined experimentally and represents the frequency of molecular collisions with the correct orientation).

Using this approach, plots of $\ln[-\ln(1 - \alpha)]$ and $\ln(\tau - \tau_i)$ were constructed from the experimental crystal phase content data. As shown in Fig. 7(a), the linearity of these plots, with correlation coefficients (R^2) greater than 0.9, confirms that the JMA model provides a reliable description of the isothermal crystallization kinetics. Notably, two distinct linear plots were obtained in the $1100^\circ\text{C} - 950^\circ\text{C}$ transition pathway of Scheme III at soaking times of $t = 10 \text{ s}$ and 20 s , respectively. This deviation reflects the sharp change in slope caused by the temperature drop. Meanwhile, the Avrami exponents shown in Fig. 7(b),

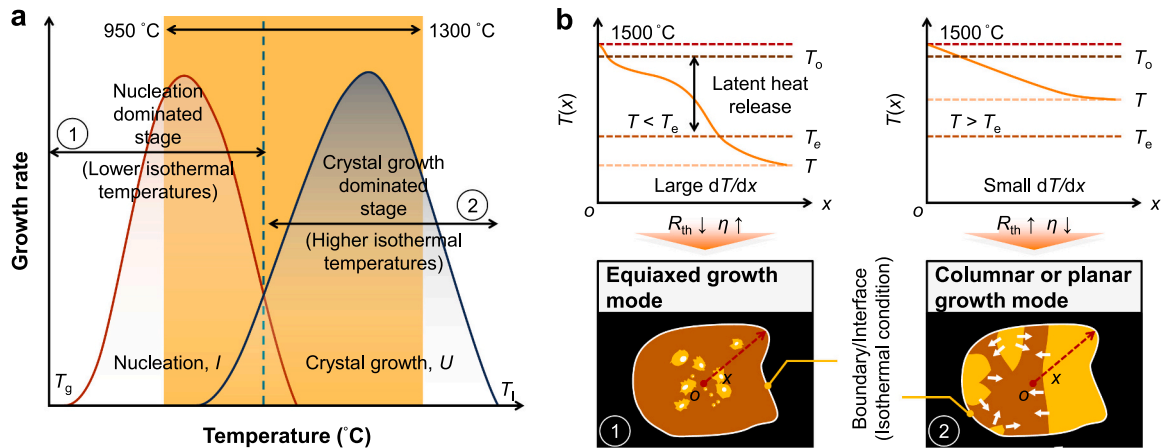


Fig. 6. Theoretical illustration of dynamic microstructure evolution. (a) Superposition of steady-state nucleation rate (I) and crystal growth rate (U) as functions of temperature, based on theoretical justification. The peaks of I and U occur sequentially with increasing isothermal temperature between T_g and T_l . (b) Variation of the temperature gradient along the radial direction of the SHTT sample at initial cooling stage, which governs heat and mass transfer and ultimately determines the growth mode by modulating thermal resistance and viscosity.

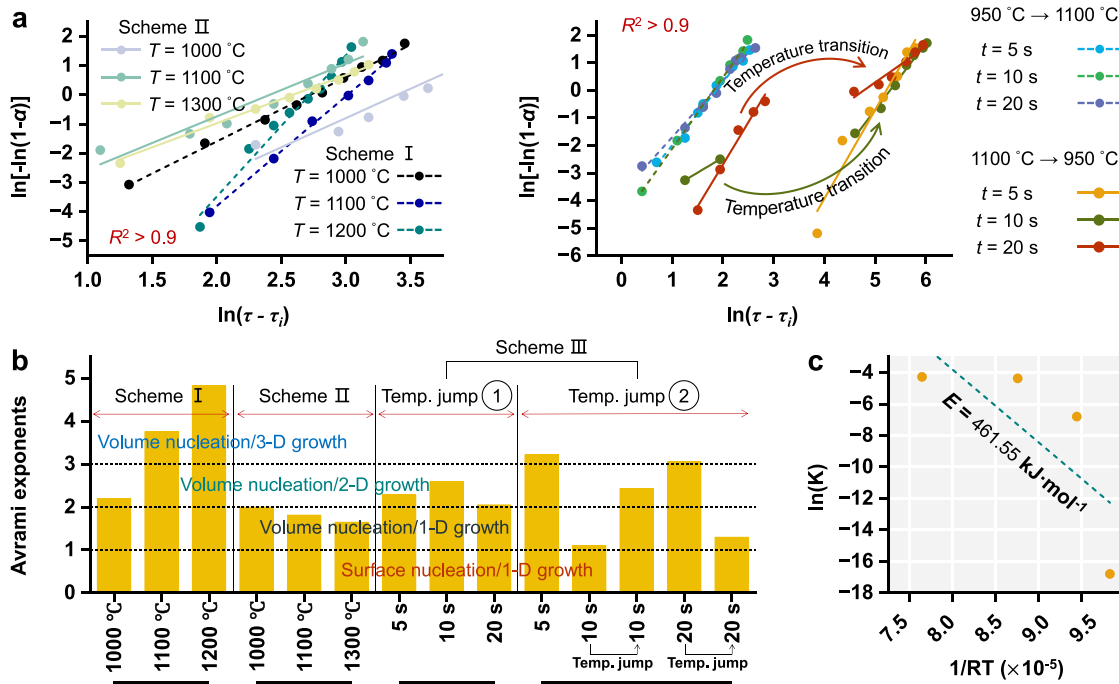


Fig. 7. Determination of crystallization kinetics parameters using isothermal JMA model. (a) Plots of $\ln[-\ln(1-\alpha)]$ versus $\ln(\tau - \tau_i)$ across the designed schemes. (b) Avrami exponents derived from the fitting results. Temperature jumps 1 and 2 represent the transition pathways of 950°C – 1100°C and 1100°C – 950°C , respectively. (c) The estimated activation energy of the simulant.

were derived from the slopes of the JMA plots. Their variation across different schemes indicates that, although similar crystallization phenomena were observed, the underlying growth mechanisms differed depending on the thermal conditions.

The results suggested that volume nucleation dominated the crystallization process across all schemes, as indicated by $n > 1$, while the growth mechanisms varied with the thermal conditions. In Scheme I, crystal growth tended toward 3D behavior as the isothermal temperature increased. By contrast, Scheme II showed a progression from 2D growth to 1D growth as temperature rose. This implies that equiaxed growth in both Scheme I and II mainly followed a 2D growth mode, whereas columnar and planar growth exhibited 3D behavior in Scheme I and 2D behavior in Scheme II. In Scheme III, different soaking times produced distinct growth modes: the 950°C – 1100°C transition pathway was characterized by 2D growth, while the reverse 1100°C – 950°C transition showed a sequence of 3D, 2D, and 1D growth modes with the increasing soaking time after the temperature drop. These shifts reflect the extensive saturation of nucleation sites prior to the onset of crystal growth under a pronounced dT/dx pathway, which favors low-dimensional growth with prolonged crystallization duration and promotes a uniform distribution of equiaxed crystals.

To further quantify crystallization kinetics, JMA plots from Scheme II and the 1100°C – 950°C transition pathway at $t = 5\text{ s}$ in Scheme III were selected for Arrhenius fitting of $\ln K$ versus $1/RT$. These conditions were chosen because they represented post-incubation stages under transitioned isothermal temperatures and spanned the full range of isothermal conditions studied. From this analysis, E was estimated to be $461.55\text{ kJ}\cdot\text{mol}^{-1}$, representing the minimum energy barrier that must be overcome for crystal precipitation, as shown in Fig. 7(c).

3.3. Implementation scenario and perspectives

The pursuit of establishing lunar bases represents a gradual and challenging process that necessitates continuous technological innovation and systems integration. Once accomplished, it would transform space exploration from a distant aspiration into a practical endeavor,

while alleviating pressure on terrestrial resources and environmental resilience. Furthermore, establishing a self-sustaining lunar resource system could form the basis of an extraterrestrial-terrestrial resource cycle, supporting the transition to low-carbon, renewable energy systems on Earth. In the early stages of lunar base construction, infrastructure is expected to rely primarily on Earth-supplied materials. However, for long-term expansion, ISRU of lunar soil for producing glass-ceramics becomes essential.

Herein, we propose an aspirational strategy for producing lunar soil-based glass-ceramics as construction materials for future lunar bases, as illustrated in Fig. 8(a). Once essential equipment has been transported to a designated site (preferably on a mare plain), the primary challenge becomes securing a reliable energy supply. Solar energy, which is abundant and readily available on the Moon, can be harnessed to power the entire system. The first step in glass-ceramic production is verifying lunar soil to obtain parent glass, an energy-intensive process. This can be accomplished by concentrating sunlight with concave mirrors [42], eliminating the need for complex melting equipment. In parallel, lunar rovers would collect raw materials and continuously feed them into the melting unit, enabling a stable supply of molten glass for downstream manufacturing processes, where the operating conditions could be configured to approximate terrestrial atmospheric environments when applicable.

During the subsequent thermal treatment of the parent glass, it is crucial to achieve crystals of uniform size and orientation to ensure the long-term durability of glass-ceramics under the harsh lunar environment. Insights from the microscale investigation of lunar soil simulant crystallization suggest that increasing the degree of supercooling during thermal treatment is an effective cooling strategy for precisely controlling crystallization behavior and thus improving material quality. As illustrated in Fig. 8(b), strong supercooling (large dT/dx) favors I over U , resulting in a higher number of finer crystals. This suppresses excessively rapid U , which would otherwise generate large, heterogeneously distributed coarse grains and compromise structural integrity shown in Fig. 8(c), because the reduced degree of supercooling weakens the driving force for nucleation and intensifies the associated heat- and

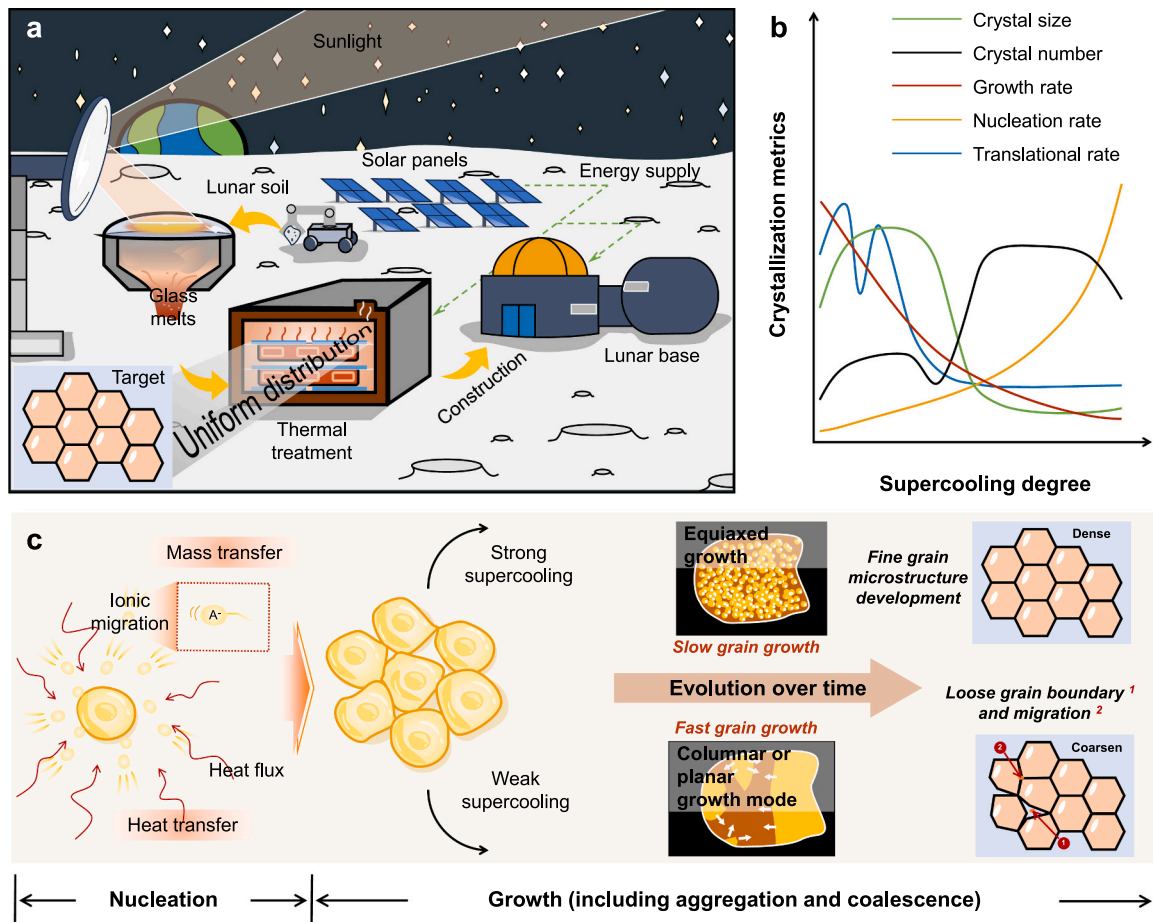


Fig. 8. Overview of *in-situ* lunar soil-based glass-ceramic fabrication. (a) Conceptual visualization of *in-situ* lunar soil-based glass-ceramic material production for base construction on the Moon. (b) Control strategies for crystallization behavior, illustrated using experimental results. (c) Schematic interpretation of the fundamental mechanisms governing crystal grain development under varying degrees of supercooling. Higher supercooling promotes more uniform grain distribution and enhanced structural densification.

mass-transfer processes for growth. With the I still finite, a sufficiently high U at the onset of nucleation results in pronounced growth of the nuclei formed during cooling, a kinetic condition that is typically associated with low cooling rates and is generally undesirable for glass-ceramic preparation. Effectively, the formation of numerous small and uniformly distributed crystals enables cooperative stress resistance, helping prevent cracking and enhancing the mechanical stability of the glass-ceramics, which can be applied in the scenario of walls, roads, and windows to maintain safety.

To advance this vision, several key perspectives should be considered. Achieving the desired hybrid microstructure requires controlled crystallization within the parent glass. A central challenge lies in achieving predictive control over crystallization, which necessitates the development of models that quantitatively link key processing parameters (e.g., cooling rate, supercooling degree, and temperature) to crystallization behavior and the resulting material properties, while the incorporation of dimensionless compositional ratios (e.g., CaO/SiO_2 , $\text{MgO}/\text{Al}_2\text{O}_3$, and $\text{CaO}/\text{Al}_2\text{O}_3$) could further enhance the generality of these models across diverse feedstocks. Meanwhile, vacuum and microgravity are the two primary environmental factors expected to influence crystallization behavior, while other factors (e.g., temperature fluctuations and radiation) can be largely mitigated within controlled habitats. Our study supported the negligible effect of gravity via the low Gr value, primarily due to the small sample mass ($\sim 10^{-3}$ g). Building on this, the proposed models could be validated through experiments under simulated vacuum conditions or, ideally, in space-based platforms. Such efforts would enable further refinement of the models and enhance their

reliability for practical deployment.

4. Conclusion

Using a prepared lunar soil simulant and performing microscale crystallization analysis under three designed thermal schemes via the SHTT method, we show that crystallization behavior exhibits strong dependence on thermal conditions, with increasing isothermal temperature ($1000\text{--}1200^\circ\text{C}$) driving a transition from equiaxed to columnar and ultimately planar growth, while mixed growth modes emerge under temperature-jump schemes at lower temperatures. This evolution reflects the competing roles between nucleation and growth, where lower temperatures promote nucleation and higher temperatures enhance growth, leading to substantial variations in crystallization duration (up to 28-fold). Correspondingly, planar growth achieved the highest growth rates, whereas equiaxed growth produced fine and uniformly distributed grains, with volume nucleation dominating across all conditions ($n > 1$) and an activation energy of $461.55\text{ kJ}\cdot\text{mol}^{-1}$.

Thermal schemes involving temperature transitions fundamentally alter dT/dx , which controls the supercooling effect that governs the dynamic balance and competition between nucleation and crystal growth. Our findings suggest that tailoring supercooling during thermal processing is key to controlling microstructure, providing a viable strategy that leverages strong supercooling to produce lunar soil-derived glass-ceramics with uniform and well-aligned crystal architectures for extraterrestrial construction applications. The underlying crystallization mechanisms and trends identified here are expected to be applicable to

other compositions with similar compositional ratios (i.e., $\text{MgO}/\text{Al}_2\text{O}_3 = 2.54$ and $\text{CaO}/\text{Al}_2\text{O}_3 = 0.54$).

Despite these insights, this study also outlines several perspectives on potential implementation scenario to enlighten future work. Addressing these aspects will advance the understanding of both the potential and challenges of utilizing lunar soil as a raw material for glass-ceramics, thereby supporting *in-situ* resource utilization, promoting technological innovation in material recycling, all of which contribute to efficient resource management and sustainable extraterrestrial-terrestrial resource cycle.

CRedit authorship contribution statement

Xudong Chen: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Qiqi Han:** Methodology, Investigation, Data curation. **Yucheng Qiao:** Resources, Investigation. **Yu Tan:** Resources, Investigation. **Zhang Bai:** Resources, Investigation. **Bin Ding:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.ijheatmasstransfer.2026.128955](https://doi.org/10.1016/j.ijheatmasstransfer.2026.128955).

Data availability

Data will be made available upon reasonable request.

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